MILITARY STANDARD

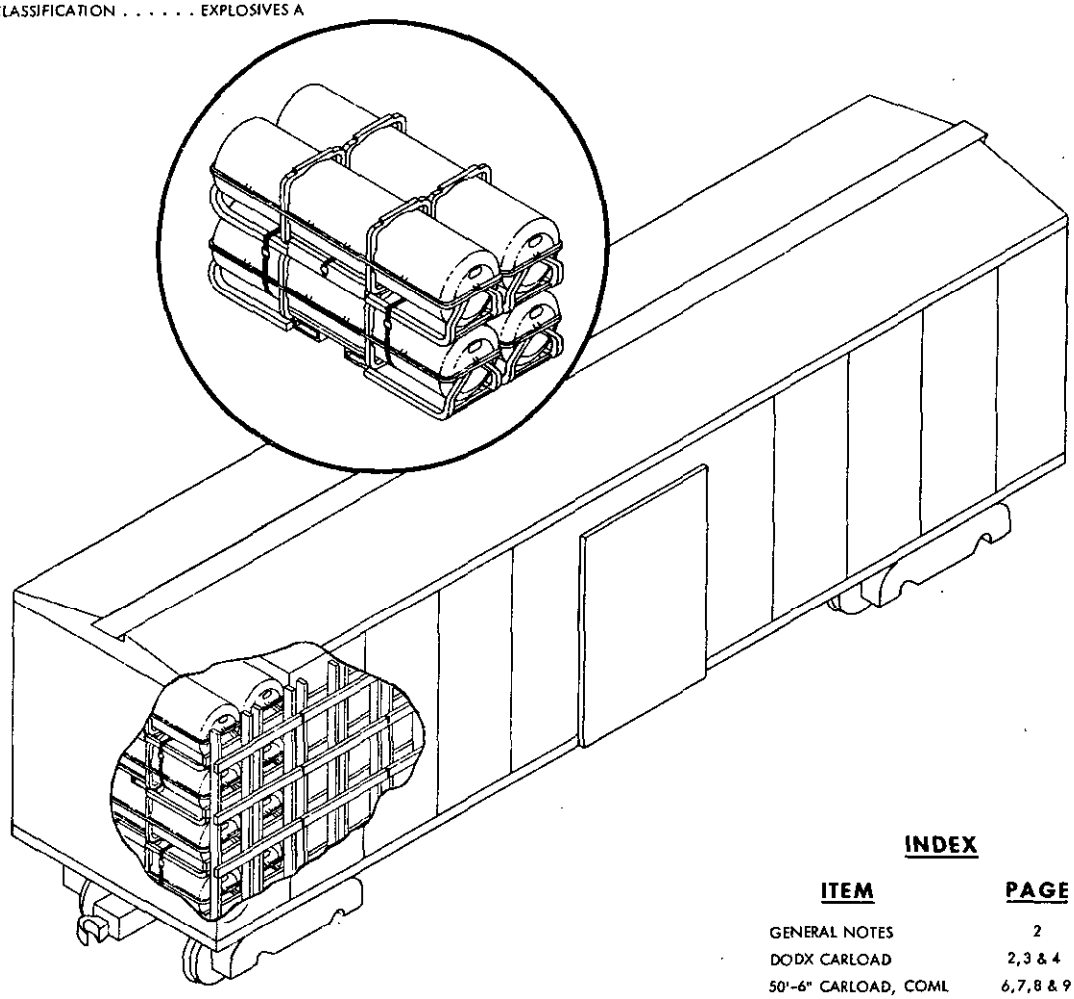
MIL-STD-1325-161
(NAVY)

10 APRIL 1973

CARLOADING
PROPULSION SECTION, (PHOENIX)
IN CONTAINER CNU-159/E
DOMESTIC UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-53/838
WEIGHT . . . . . 3380 LBS
DIMENSIONS . . . . . 95 1/8 L x 46 5/16 W x 49 7/16 H
STACKING HEIGHT 49
CUBE . . . . . 126.0 CU FT
DOT HAZARD CLASSIFICATION . . . . . EXPLOSIVES A



INDEX

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GENERAL NOTES	2
DODX CARLOAD	2,3 & 4
50'-6" CARLOAD, COML	6,7,8 & 9

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

AUTHORIZED AND RELEASED FOR GENERAL USE.

APPROVED BY BUREAU OF EXPLOSIVES
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SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF PROPULSION SECTIONS (PHOENIX) IN CONTAINERS CNU-159/E. THE CONTAINERS ARE UNITIZED FOUR TO A UNIT LOAD AS SHOWN IN WR-53/838.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. APPLICABLE MATERIAL SPECIFICATIONS:

   DUNNAGE LUMBER - FED. SPEC. MM-L-751

   NAILS - FED. SPEC. FF-N-105

   STRAPPING - FED. SPEC. QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).

   SEALS - FED. SPEC QQ-S-781, STYLE III. HEAVY DUTY
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL. 1).

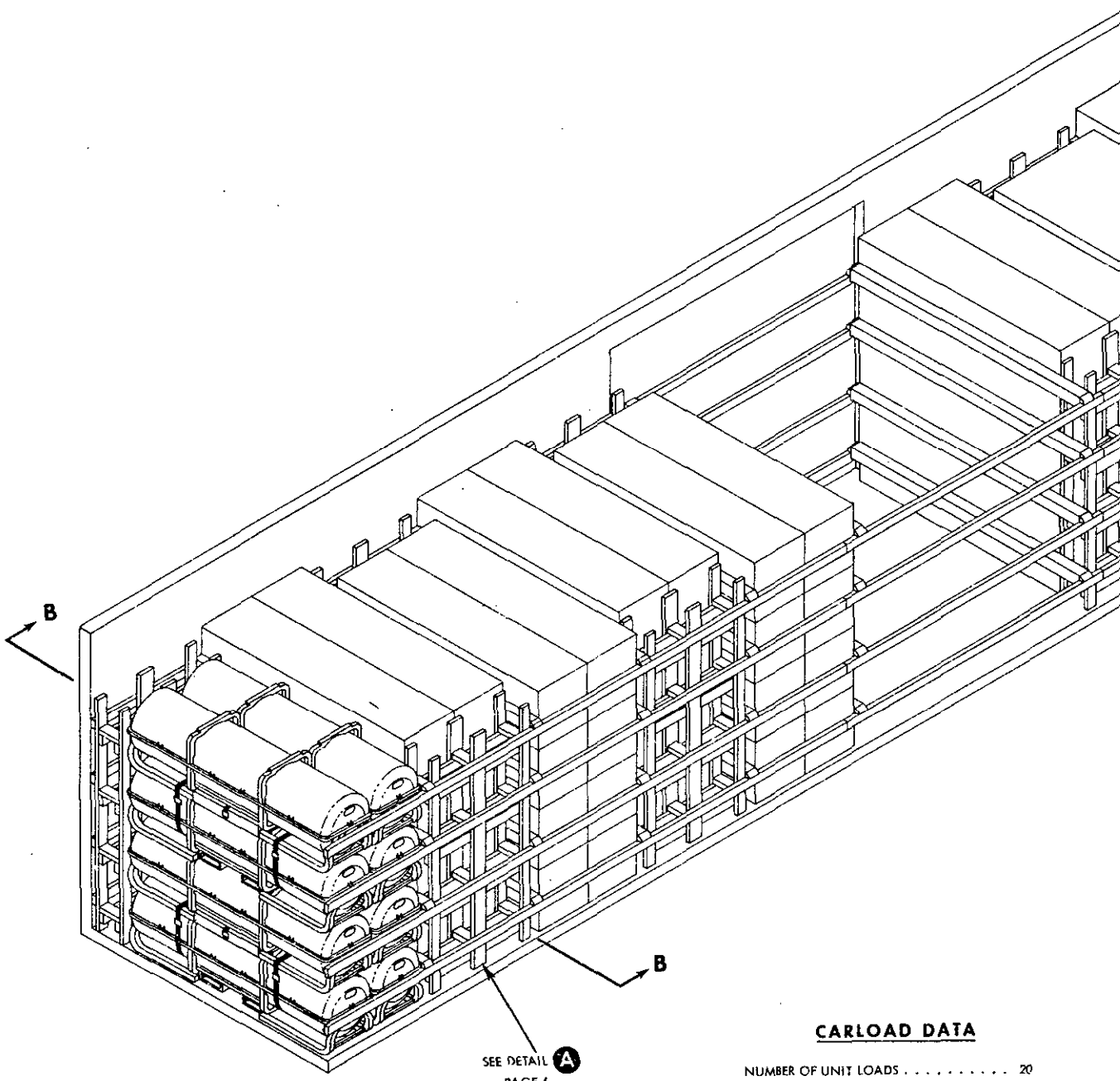
50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 20 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH. SEE MIL-STD-1325(NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS: NUMBER	NAILS: SIZE
4	CROSS BRACE	2 x 4 x CUT TO FIT	120	SEE 1, 3	-	-
3	SWAY BRACE VERTICAL	2 x 6 x 96	20	4	2 PER JOINT	16d
2	SWAY BRACE HORIZONTAL	2 x 4 x 46	80	1,3	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 x 4 x 96	40	4	2 PER JOINT	16d

LIST OF MATERIALS & NAILING DATA

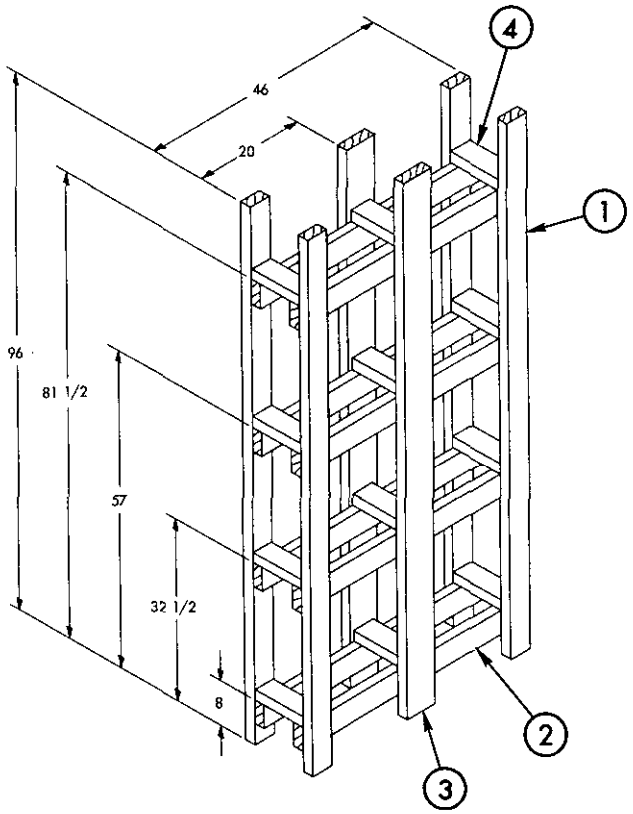
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CARLOAD DATA

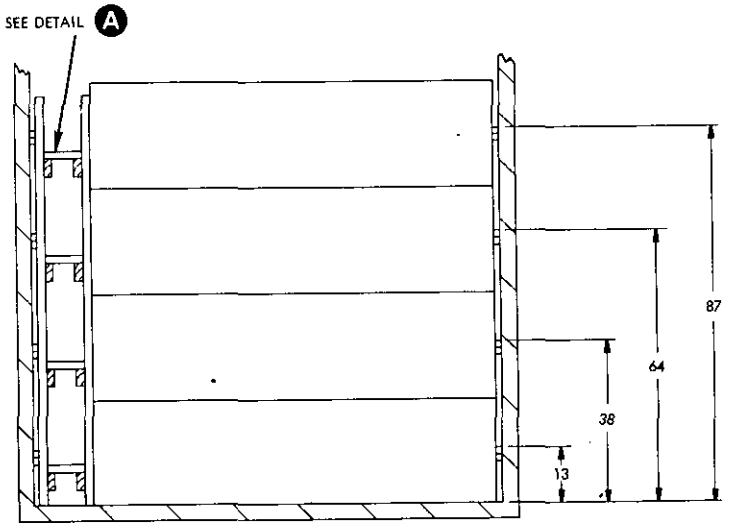
NUMBER OF UNIT LOADS	20
NUMBER OF CROSSMEMBERS REQD	40
LOAD WEIGHT	67,600 LBS
DUNNAGE WEIGHT	1,655 LBS
CARLOAD WEIGHT	69,255 LBS

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DETAIL A

SWAY BRACE ASSEMBLY
10 REQUIRED



SECTION B-B

SHOWING LOCATION OF WALL MEMBERS AND DOORWAY MEMBERS
USED FOR CROSSMEMBERS

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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 24 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 16 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 16 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's

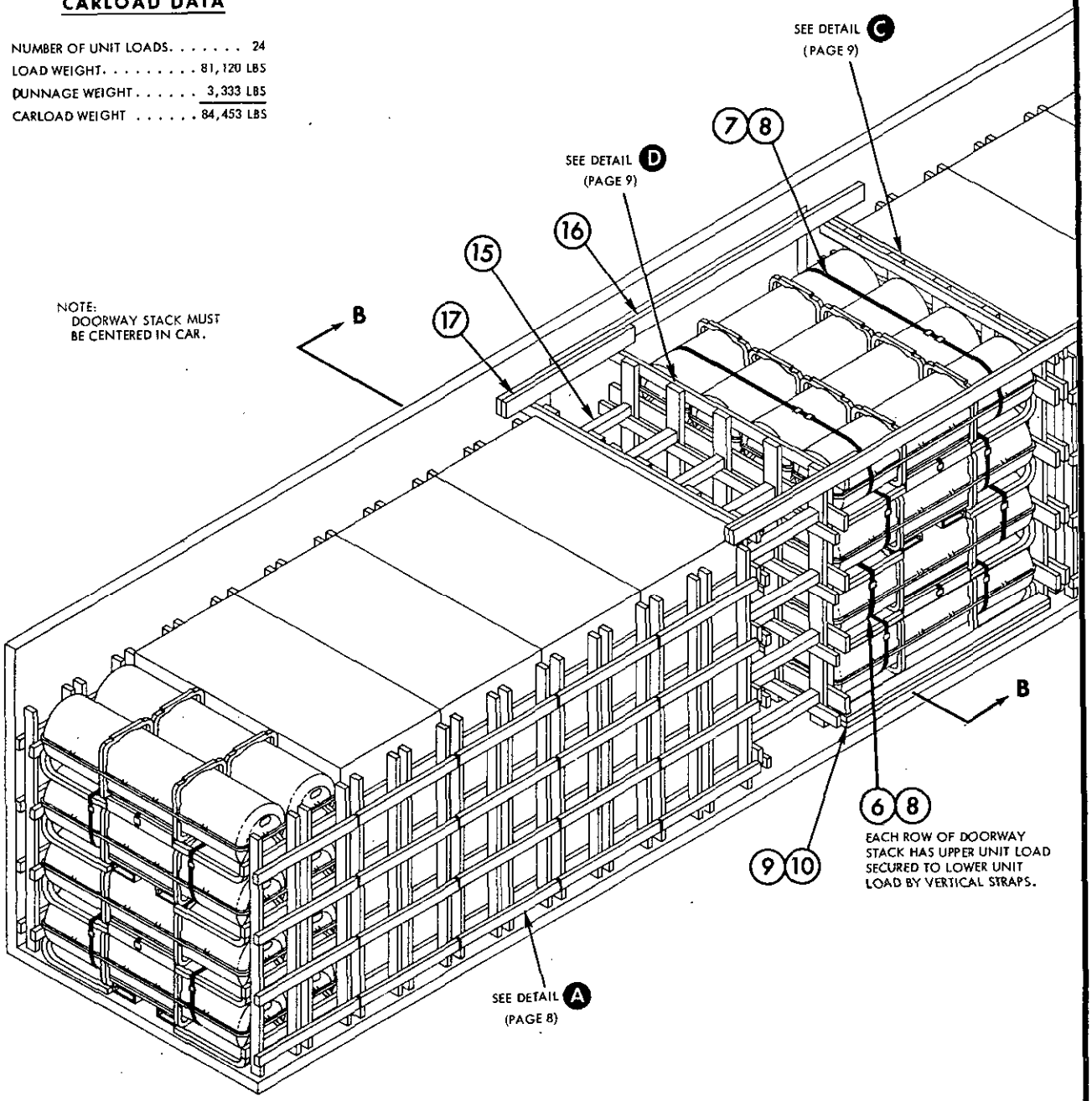
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
17	HOLD DOWN CLEAT	2 x 6 x 60	2	16	2 PER FOOT	10d
16	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
15	STRUT	4 x 4 x WEDGE FIT*	20	12	2 PER JOINT	16d
14	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	2	12	3 PER JOINT	10d
13	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	12	3 PER JOINT	10d
12	CENTER GATE VERTICAL	2 x 6 x 102	10	SEE 11	-	-
11	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	12	3 PER JOINT	10d
10	SLEEPER	2 x 6 x 96	2	9	2 PER FOOT	16d
9	SLEEPER	2 x 6 x 96	2	CAR FLOOR	2 PER FOOT	16d
8	SEAL	1 1/4	12	-	-	-
7	CROSS STRAP	1 1/4 x .035 x 20 FT 6	2	-	-	-
6	VERTICAL STRAP	1 1/4 x .035 x 12 FT 6	4	-	-	-
5	TOP HORIZONTAL	2 x 4 x CAR WIDTH-1	1	3	3 PER JOINT	10d
4	SEPARATOR GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	3	3 PER JOINT	10d
3	SEPARATOR GATE VERTICAL	2 x 6 x 102	5	SEE 4	-	-
2	SWAY BRACE HORIZONTAL	2 x 4 x 46 1/4	160	1	2 PER JOINT	16d
1	SWAY BRACE VERTICAL	2 x 4 x 96	80	SEE 2	-	-

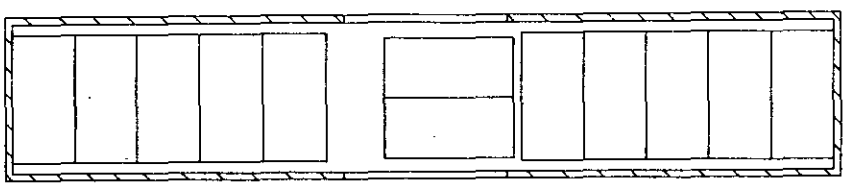
LIST OF MATERIALS & NAILING DATA

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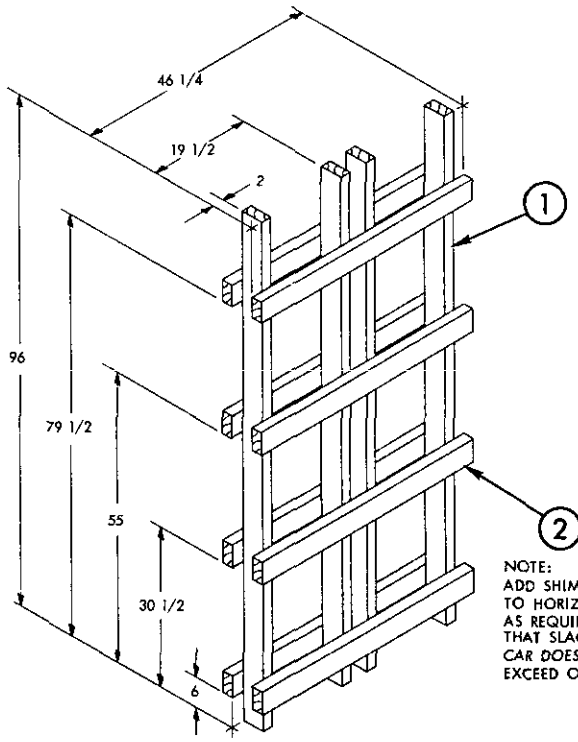
NUMBER OF UNIT LOADS	24
LOAD WEIGHT	81,120 LBS
DUNNAGE WEIGHT	3,333 LBS
CARLOAD WEIGHT	84,453 LBS





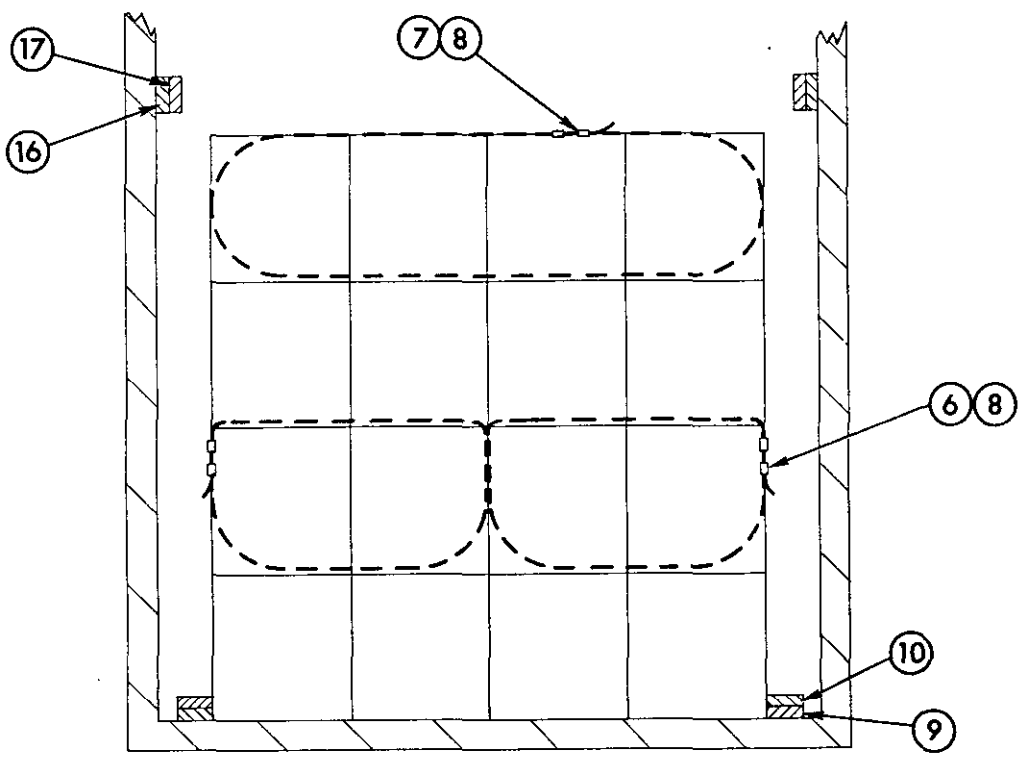
LOADING PLAN

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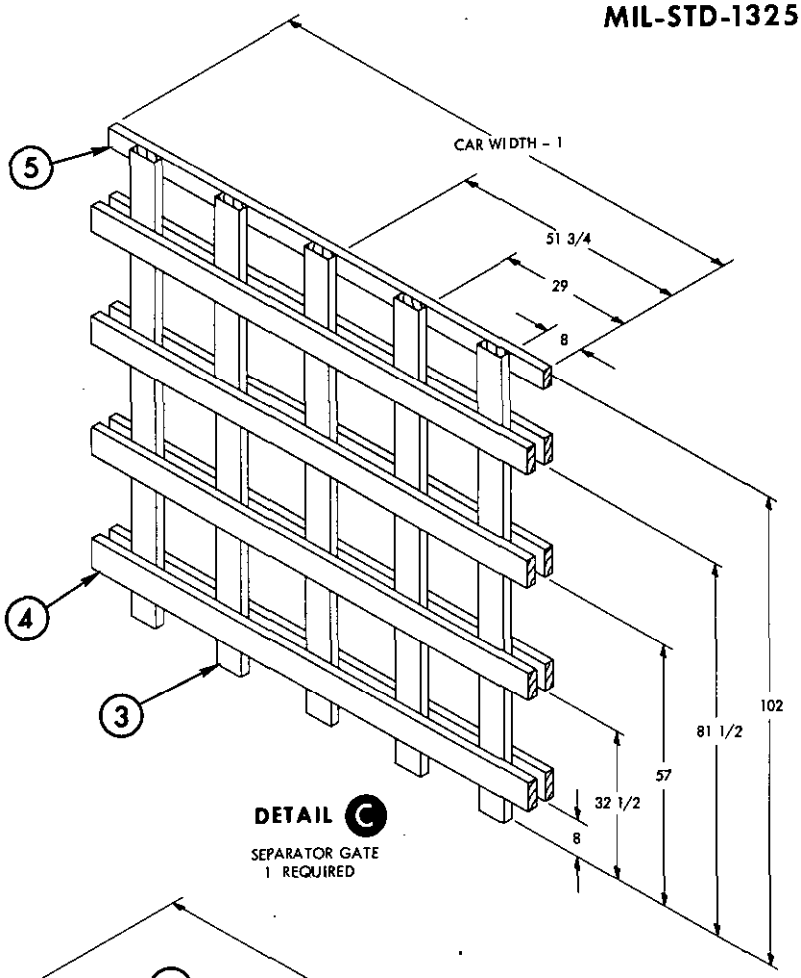
DETAIL A

SWAY BRACE ASSEMBLY
20 REQUIRED



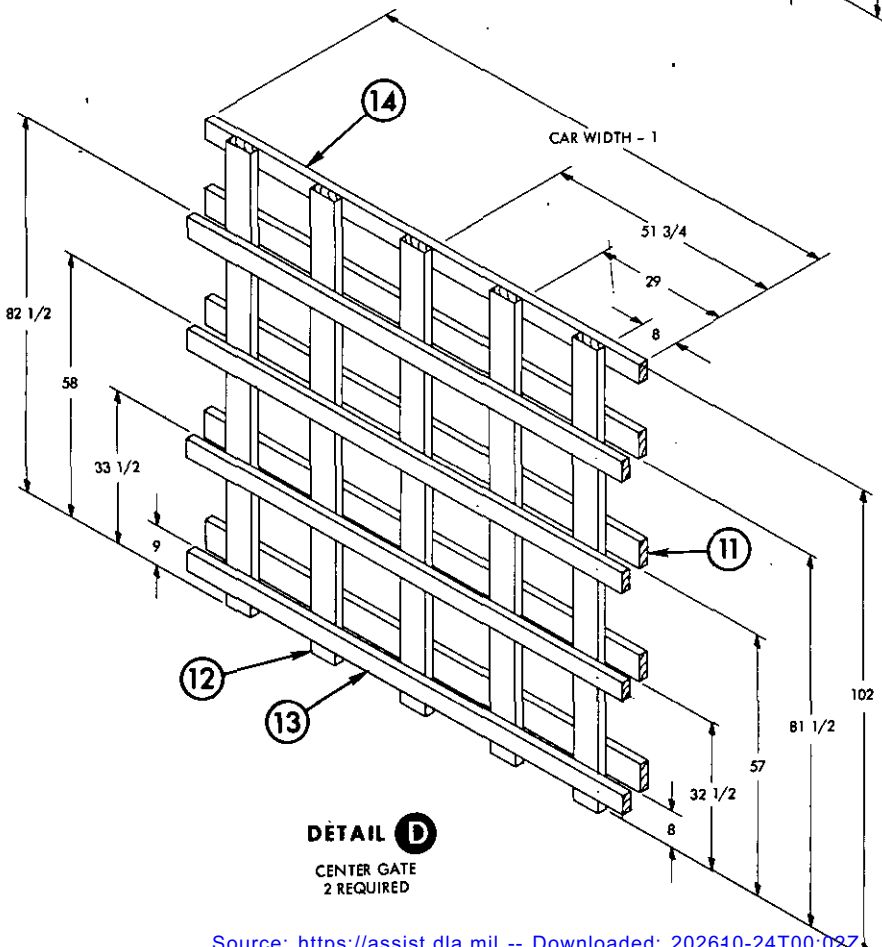
SECTION B-B

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DETAIL C

SEPARATOR GATE
1 REQUIRED



DETAIL D

CENTER GATE
2 REQUIRED

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